Industrial Internship Report on

"Content Management System (CMS) Blog Website"

Prepared by

Mohasin Khan

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and UniConverge Technologies Pvt Ltd (UCT). This internship was focused on developing a Content Management System (CMS) Blog Website. The project was completed within the given duration. My project was to design and develop a CMS-based blog website using HTML, CSS, JavaScript, PHP, and MySQL. This internship gave me a great opportunity to understand real-world development, improve my coding skills, and gain practical experience in full stack development.

TABLE OF CONTENTS

1 Preface 3

2 Introduction 4

2.1 About UniConverge Technologies Pvt Ltd 4

2.2 About upskill Campus 8

2.3 Objective 9

2.4 Reference 9

2.5 Glossary 10

3 Problem Statement 11

4 Existing and Proposed solution 12

5 Proposed Design/ Model 13

5.1 High Level Diagram (if applicable) 13

5.2 Low Level Diagram (if applicable) 13

5.3 Interfaces (if applicable) 13

6 Performance Test 14

6.1 Test Plan/ Test Cases 14

6.2 Test Procedure 14

6.3 Performance Outcome 14

7 My learnings 15

8 Future work scope 16

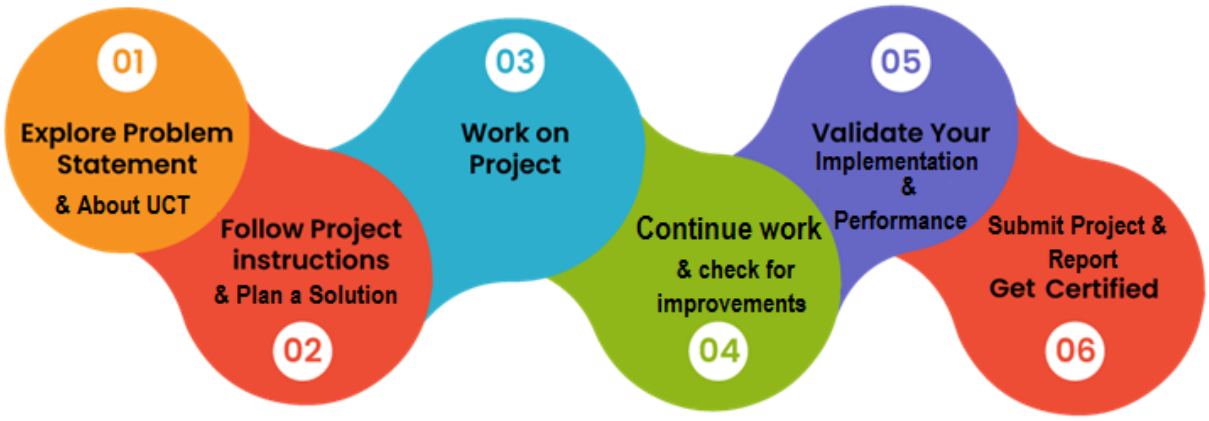
1 Preface

This report presents the summary of my 6-week Full Stack Development Internship conducted by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT).

This internship was very important for my career development as it provided practical exposure to real-world projects and improved my technical skills. During this internship, I worked on a project titled "Content Management System (CMS) Blog Website", where I developed a simple platform for adding and displaying blog posts.

The program was well planned with weekly learning modules, quizzes, and project-based tasks. It helped me understand both front-end and backend development step by step.

Through this internship, I learned HTML, CSS, Bootstrap, PHP, and MySQL, and understood how to build a complete full stack web application. This experience improved my problem-solving skills and boosted my confidence in web development.



I would like to express my sincere thanks to upskill Campus, UniConverge Technologies Pvt Ltd, and all the mentors who guided and supported me throughout this internship.

I would also like to thank my teachers and friends who helped me directly and indirectly during this project.

Finally, I would like to suggest my juniors and peers to participate in such internships as they provide valuable practical knowledge and help in building a strong foundation for future careers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

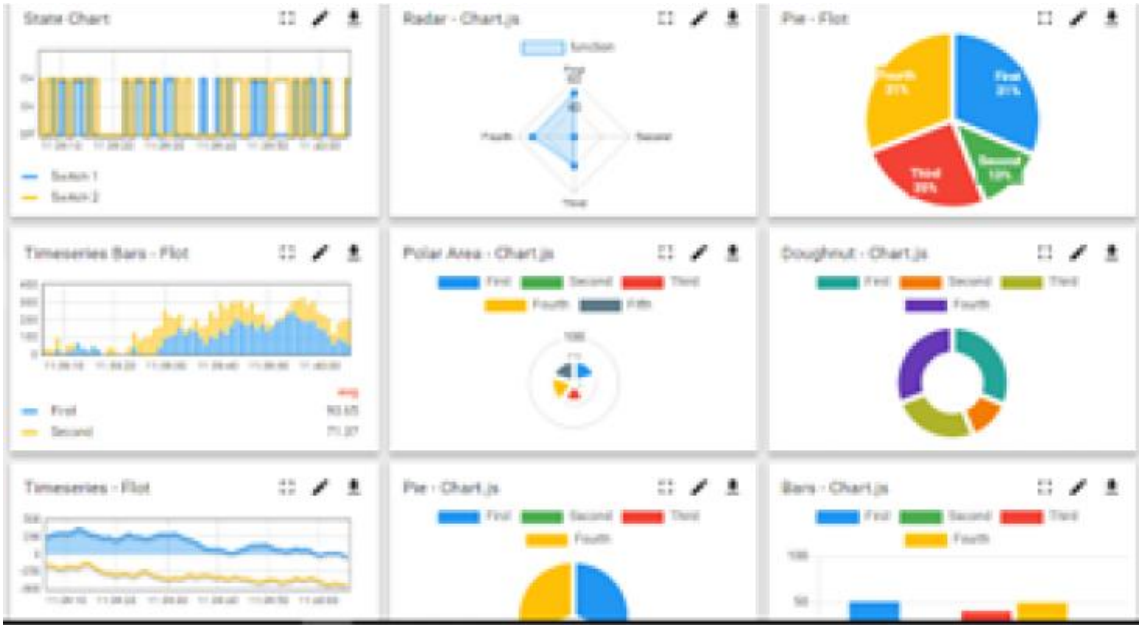
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

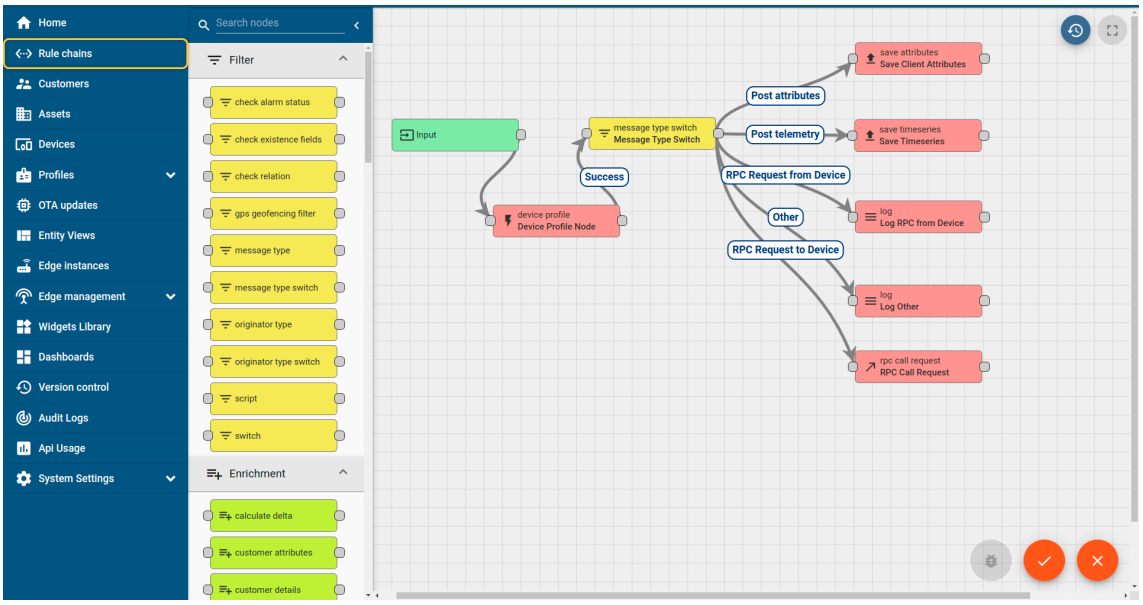
- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

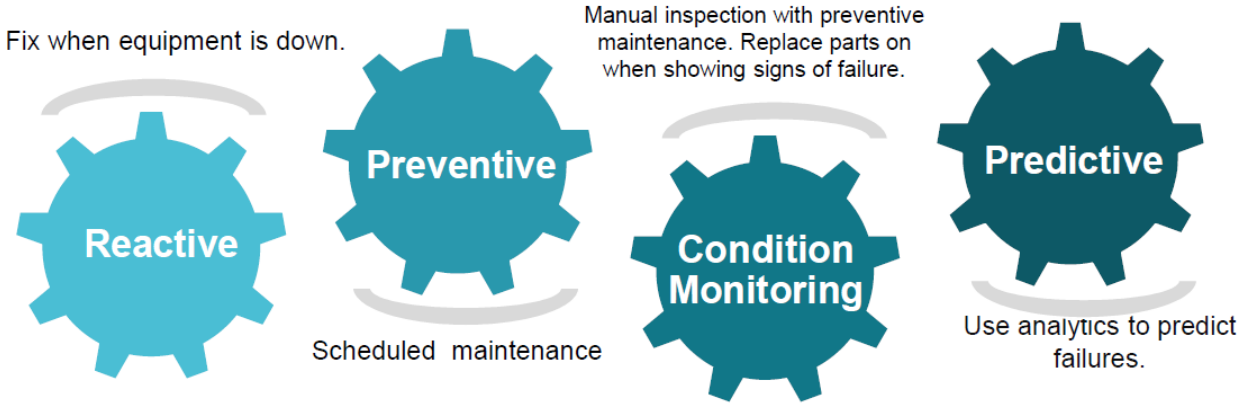
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

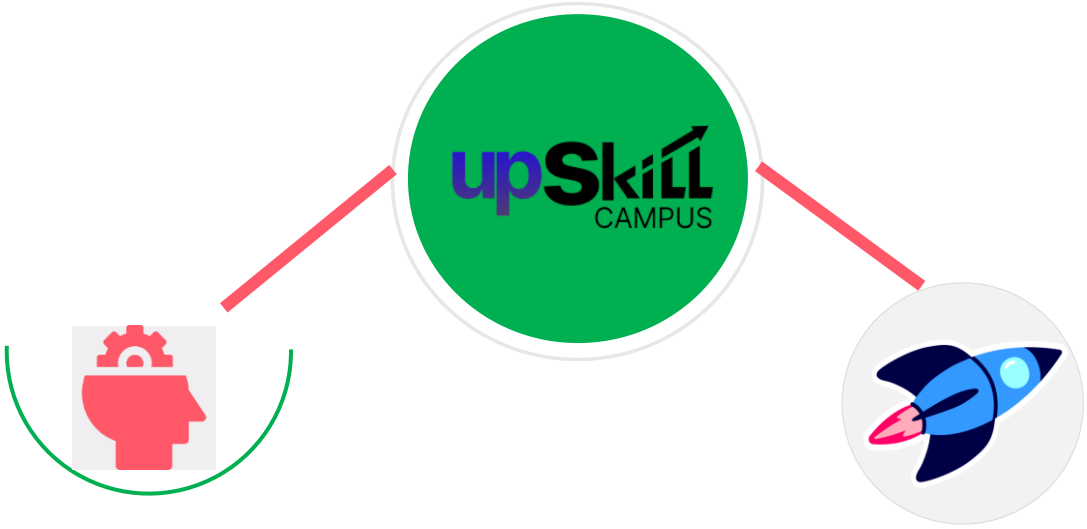
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

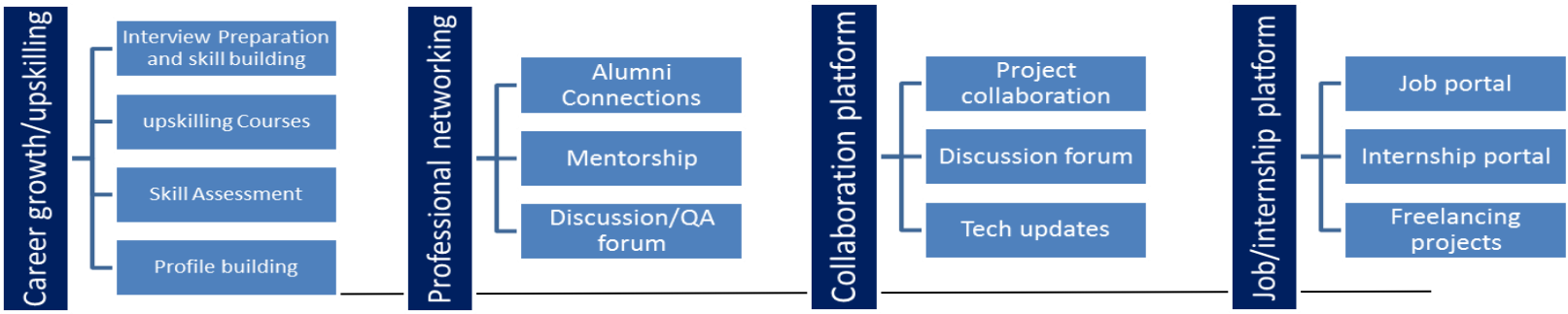
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Chatgpt

[2] Microsoft clipbord

[3] Ms. Word Documentation

2.6 Glossary

Terms	Acronym

Contant Management System	CMS
Hyper Text Markup Language	HTML
Cascading Style Sheets	CSS
Hypertext Preprocessor	PHP
Database Management System	DBMS

3 Problem Statement

In the assigned problem statement

In this project, the main problem was creating a simple and user-friendly platform where users can easily write and manage blog posts.

Many existing blogging systems are complex and require technical knowledge, which makes it difficult for beginners to use them. There was a need for a basic Content Management System (CMS) that allows users to add and view blog posts without any difficulty.

To solve this problem, I developed a CMS Blog Website using HTML, CSS, Bootstrap, PHP, and MySQL. This system provides a simple interface where users can add blog posts and view them on the homepage easily.

This project focuses on simplicity, ease of use, and basic functionality for learning and practical implementation.

4 Existing and Proposed solution

There are many blogging platforms available, but most of them are complex and not beginner friendly. Some platforms require login systems and advanced setup, which makes it difficult for new users.

These systems are not simple, require more configuration, and are not suitable for basic learning projects.

To solve this problem, I developed a simple CMS Blog Website using HTML, CSS, PHP, and MySQL. This system allows users to add and view blog posts easily.

The main advantage of my project is simplicity and ease of use. It is lightweight, user-friendly, and helpful for beginners to understand full stack development.

4.1 Code submission (GitHub link):

https://github.com/Mohasin-Khan-Developer1/Upskillcampus.git

4.2 Report submission (Github link):

4.3 https://raw.githubusercontent.com/Mohasin-Khan-Developer1/Upskillcampus/main/Mohasin_CMS_Project_Report.pdf

5 Proposed Design/ Model

The system is designed as a simple CMS Blog Website where users can add blog posts and view them on the homepage.

Flow:

User → Add Post Page → Data stored in Database → Display on Homepage

5.1 High Level Diagram (if applicable)

User → Website (Frontend) → Server (PHP) → Database (MySQL)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

User Input → Form (Add Post Page) → PHP Processing → Database Storage → Fetch Data → Display on Homepage

5.3 Interfaces (if applicable)

The system includes:

- Homepage (to display blog posts)

- Add Post Page (to insert new blog data)

Database connection interface using PHP

6 Performance Test

The system was tested for basic performance such as loading speed, data storage, and retrieval.

Constraints:

- Limited features

- Basic UI

- Small database

The system performs well for small-scale usage and learning purposes.

6.1 Test Plan/ Test Cases:

6.2 Test adding a blog post
6.3 Test displaying blog post
6.4 Test database connection

6.5 Test Procedure:

6.6 Open website
6.7 Go to Add Post page
6.8 Enter data and submit
6.9 Check homepage for output

6.10 Performance Outcome:

The system worked successfully. Blog posts were added and displayed correctly without errors.

7 My learnings

During this internship, I gained practical knowledge of full stack web development. I learned how to design web pages using HTML, CSS, and Bootstrap, and how to connect frontend with backend using PHP and MySQL.

I understand how to create a complete CMS Blog Website where users can add and view blog posts. I also learned about database handling, form processing, and dynamic data display.

This internship improved my problem-solving skills and helped me understand real-world project development. It also increased my confidence in building web applications and working with different technologies.

8 Future work scope

In the future, this project can be improved by adding more advanced features such as user authentication (login/signup system), edit and delete post options, and comment functionality.

The design of the website can be further enhanced to make it more attractive and fully responsive for all devices. Security features can also be added to protect user data.

Additionally, the project can be deployed on a live server so that it can be accessed online. These improvements will make the system more useful and closer to real-world applications.